An exact two-charge formula for the memory core
Charged covariance transfer and its double-scaled filter

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Abstract

A charged-covariance transfer identity reduces the full quadratic Gaussian contraction of the quartic memory core to two coherent products in adjacent charge sectors. The resulting finite-N coefficients are explicit, including the channels forbidden to ordinary multiplication at half filling. After an exact cancellation, the H^2 coefficient has a linear rank-deficit limit. Existing positive product tails promote this part to an annealed L^2 approximation. The weighted A part and the output-deficit cross spectrum remain the next steps toward the full memory observable.

1 Conventions and the transfer identity

Use standard CAR and the overlapping complex Gaussian p-body ensemble, $\mathbb{E}J_{IJ}J_{KL} = w\delta_{IL}\delta_{JK}$, $\hbar = 1$. The state is maximally mixed at fixed charge, with normalized trace tr_m and $\|X\|_2^2 = \mathbb{E}\,\mathrm{tr}_m X^\dagger X$. Set $h = H_m$, $k = H_{m-1}$, $c_i : \mathcal{H}_m \to \mathcal{H}_{m-1}$, and $\mathrm{Rem}\, X = \sum_i c_i^\dagger X c_i$. All finite formulas below assume $m > p \geq 1$; the double-scaled regime satisfies this condition. Write $d = m - p$, $n = N - m + 1$, and

$$C = w\binom{m}{p}\binom{N-m+p}{p}, \qquad C_- = w\binom{m-1}{p}\binom{N-m+p+1}{p}, \tag{1}$$

$$j_c = w\binom{m-1}{p}\binom{N-m+p}{p}, \qquad \kappa_r = w\binom{m-r}{p}\binom{N-m+p-r}{p}, \tag{2}$$

$$e_r = \frac{w}{p}\binom{m-r-1}{p-1}\binom{N-m+p-r}{p-1}. \tag{3}$$

Out-of-range binomials are zero. In the interior, $e_r = p\kappa_r/[(m-r)(n-r)]$.

For an independent copy (u, v) sharing the same new couplings in the two sectors, define $\Psi(X) = \mathbb{E}uXv$. For arbitrary rectangular matrices $Y_i : \mathcal{H}_{m-1} \to \mathcal{H}_m$, the key identity is

$$\boxed{P_r \sum_i \Psi(Y_i)c_i = \kappa_r P_r \sum_i Y_i c_i - e_r P_r\, \mathrm{Rem}\left(\sum_i c_i Y_i\right).} \tag{4}$$

No rank restriction on Y_i is required. The output rank is $0 \leq r \leq \min(m, N-m)$.

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2 Why only two charged components are needed

Let $T_r^{\pm}$ be the same primitive traceless rank tensor restricted to charges m and $m-1$. Introduce vector-valued operators

$$V_i = T_r^+ c_i^\dagger, \qquad U_i = c_i^\dagger T_r^-. \tag{5}$$

Their difference is a commutator with one creation operator; it is the traceless charged tensor of type $(r, r-1)$. The remaining component has type $(r+1, r)$. Normal ordering with a single extra creation operator permits precisely these two types. Their known embedded-covariance eigenvalues are

$$L_r = w\binom{m-r}{p}\binom{N-m+p-r+1}{p}, \tag{6}$$

$$U_r^{\mathrm{eig}} = w\binom{m-r-1}{p}\binom{N-m+p-r}{p}. \tag{7}$$

Put $D_r = N+1-2r$. Then

$$L_r - U_r^{\mathrm{eig}} = D_r e_r, \qquad U_r^{\mathrm{eig}} + (n-r)e_r = \kappa_r. \tag{8}$$

Use unnormalized HS norms in this paragraph, and set $A_+ = \|T_r^+\|^2$, $A_- = \|T_r^-\|^2$. Removal and addition give

$$\frac{A_-}{A_+} = \frac{m-r}{n-r}, \quad \|V-U\|^2 = \frac{rD_r}{n-r}A_+, \quad \langle V, V-U\rangle = rA_+. \tag{9}$$

Thus the low component of V is $(n-r)(V-U)/D_r$, and

$$\Psi(V_i) = \kappa_r V_i - (n-r)e_r U_i. \tag{10}$$

At $r=0$ the difference is zero; endpoints follow by the surviving component or the binomial formula.

Define $\mathsf{S}(Y) = \sum_i Y_i c_i$ and $\mathsf{T}(Y) = \sum_i c_i Y_i$. Their HS adjoints are $\mathsf{S}^*(X)_i = X c_i^\dagger$ and $\mathsf{T}^*(Z)_i = c_i^\dagger Z$. Also $\sum_i c_i T_r^+ c_i^\dagger = (n-r)T_r^-$. Taking the adjoint of the preceding eigenvector identity, using self-adjointness of Ψ, proves Eq. (4) on every Y, including all product multiplicities.

For the embedded Hamiltonian ranks h_a, k_a, the useful consequences are

$$\Psi(h_a c_i^\dagger) = \kappa_a h_a c_i^\dagger - \frac{p\kappa_a}{d} c_i^\dagger k_a, \tag{11}$$

$$\Psi(c_i^\dagger k_a) = -d e_a h_a c_i^\dagger + [\kappa_a + (m-n)e_a] c_i^\dagger k_a, \tag{12}$$

$$\kappa_a(m-1) = \kappa_a + (m-n-p)e_a. \tag{13}$$

These preserve the same couplings across the two charges.

3 The complete finite-N quadratic core

Recall

$$\mathcal{E} = \sum_i (\mathrm{ad}_H^2\, c_i)^\dagger\, \mathrm{ad}_H^2\, c_i + \tfrac{1}{4}[H,[H,A]], \qquad Q_{\mathcal{E}} = \tfrac{1}{2}\Delta\mathcal{E}. \tag{14}$$

The old exact contraction is $Q_{\mathcal{E}} = D_4 - 2D_{13} + 5D_{22}/4 + 7D_{121}/2 + (4p-3m)D_{40}$, with D the contractions of the five raw words.

For $a < b$ use $X \odot Y = \{X, Y\}$; for $a = b$ use $X \odot X = X^2$. Set

$$Z^+_{ab;r} = P_r(h_a \odot h_b), \quad Z^-_{ab;r} = P_r\,\mathrm{Rem}(k_a \odot k_b), \quad x_a = a(N+1-a). \tag{15}$$

Then, with all sums over allowed $a \le b, r$,

$$\boxed{Q_{\mathcal{E}} = \sum_{a\le b,r} \left(g^+_{ab;r} Z^+_{ab;r} + g^-_{ab;r} Z^-_{ab;r} \right).} \tag{16}$$

The coefficients are

$$\begin{aligned} g^+ &= (8p-2m)C_- + \frac{19}{2} d j_c + (12p-9m)C \\ &\quad + m(\kappa_a+\kappa_b) + \frac{3p-m}{2}\kappa_r - \frac{11}{4} dn(e_a+e_b), \end{aligned} \tag{17}$$

$$\begin{aligned} g^- &= 3C_- + \frac{5}{2}C - 4j_c - \frac{4m+p}{4d}(\kappa_a+\kappa_b) + \frac{1}{2}\kappa_r \\ &\quad - (m-n+p)(e_a+e_b) + \left[m + 3n - p - \frac{5(2mn - x_a - x_b)}{4d} \right] e_r. \end{aligned} \tag{18}$$

No division by Z^+ occurs. The previously found $(2,2;1)$ channel at $N=6, m=3$ is consequently retained.

To see the reduction, put $v_a = \kappa_a - n e_a$ and $u_a = d(j_c + v_a)$. The old linear U is $\sum_a u_a h_a$. For its apparently new quadratic term, Eq. (4) gives

$$P_r \sum_i \Psi(h_a c_i^\dagger k_b) c_i = d\kappa_r P_r h_a h_b - \frac{e_r (m-a)(n-a)}{d} P_r\,\mathrm{Rem}(k_a k_b). \tag{19}$$

All other terms use the three consequences on the preceding page. Adding their Hermitian conjugates produces the symmetric products and yields Eqs. (17)–(18).

Gaussian centering acts after this identity: $W_2\mathcal{E} = \sum(g^+ W_2 Z^+ + g^- W_2 Z^-)$, since $W_2\mathcal{E} = Q_{\mathcal{E}} - 2\mathbb{E}\mathcal{E}$. The boundary $m = p$ is separate: $k = 0$, $\mathcal{E} = ph^4$, and $Q_{\mathcal{E}} = pD_{40}$.

4 The finite part after merging the large terms

Since $A_{ab;r} = Z^-_{ab;r} + (2p-m)Z^+_{ab;r}$, define

$$g_A = g^-, \qquad g_H = g^+ + (m-2p)g^-, \qquad Q_{\mathcal{E}} = \sum (g_H Z^+ + g_A A_{ab;r}). \tag{20}$$

Now impose exact half filling $m = N/2$, $N = p^2/\lambda$, with fixed $\lambda > 0$, and only afterwards consider small λ. For fixed nonnegative deficits,

$$a = p - u, \qquad b = p - v, \qquad r = a + b - S, \qquad q = e^{-4\lambda}. \tag{21}$$

The explicit coefficients give

$$\frac{g_A}{C} = \frac{3 - 4q + q^2}{2} + O(p^{-1}), \tag{22}$$

$$\frac{g_H}{C} = 2p(1-q)^2 + K_0 + 8\lambda(q^2 - q)(u+v) + 8\lambda q^2 S + O(p^{-1}), \tag{23}$$

$$K_0 = \lambda\left[1 - 10q + 9q^2 + 16\lambda q - 32\lambda q^2\right]. \tag{24}$$

In particular, the order-m terms cancel before the limit is taken. The remaining order-p coefficient is independent of ranks and multiplies the conserved operator $H^2 - CI$ after centering.

Here is a direct derivation of the finite term. Write $g_H/C = B_0 + B_a\kappa_a/C + B_b\kappa_b/C + B_r\kappa_r/C$ using $C_-/C = d(n+p)/(mn)$ and $j_c/C = d/m$. For fixed deficits the exact rational coefficients expand as

$$(B_0, B_a, B_b, B_r) = (2p + \lambda, -2p - 5\lambda, -2p - 5\lambda, 2p + 9\lambda) + O(p^{-1}). \tag{25}$$

For $j = tp - s$, the terminating binomial product gives

$$\frac{\kappa_j}{C} = \prod_{l=0}^{p-1} \left(1 - \frac{j}{m-l}\right)\left(1 - \frac{j}{m+p-l}\right), \tag{26}$$

$$\log \frac{\kappa_j}{C} = -4\lambda t + \frac{4\lambda s - 4\lambda^2 t^2}{p} + O(p^{-2}). \tag{27}$$

Taylor's remainder for the logarithm is uniform on any fixed deficit window. Substituting $(t,s) = (1,u), (1,v), (2, u+v+S)$ gives Eq. (23). This is a coefficient expansion; treating the entire A contribution requires an additional weighted estimate.

5 An L^2 result for the H^2 contribution

The H^2 part can already be controlled beyond fixed rank windows. For $a = \alpha p$, $b = \beta p$, $r = \rho p$, the rational coefficients have the uniform expansions

$$B_0 = 2p + \lambda + O(p^{-1}), \tag{28}$$
$$B_a = -2p + \lambda(6 - 11\alpha) + O(p^{-1}), \tag{29}$$
$$B_b = -2p + \lambda(6 - 11\beta) + O(p^{-1}), \tag{30}$$
$$B_r = 2p + \lambda(5\alpha + 5\beta + 6\rho - 13) + O(p^{-1}). \tag{31}$$

Uniformity holds for $\alpha, \beta \in [0, 1]$, $\rho \in [0, 2]$: after scaling, every rational denominator stays away from zero. The binomial product and its rank finite differences satisfy $|\kappa_{j+1}/C - \kappa_j/C| \le K_\lambda/p$ throughout this window. Comparison with $j = p, 2p$ therefore gives

$$\left| \frac{g_H}{C} - 2p(1-q)^2 \right| \le K_\lambda(1 + u + v + S). \tag{32}$$

Let $C_a I = \mathbb{E}h_a^2$, $\omega_a = C_a/C$. Different input-rank pairs are orthogonal in second Gaussian chaos, and output ranks are HS orthogonal. The previous independent-product law $\pi_{ab}^{(m)}(r)$ and symmetrization bound imply, for any real filter f,

$$\left\| C \sum_{a \le b, r} f_{ab;r} W_2 Z_{ab;r}^+ \right\|_2^2 \le 4C^4 \sum_{a,b} \omega_a \omega_b \sum_r \pi_{ab}^{(m)}(r) |f_{ab;r}|^2. \tag{33}$$

The established input-deficit factorial tail and uniform third moment of S make the square of Eq. (32) uniformly integrable. First restrict u, v, then use the $O(K^{-1})$ squared tail for S; the complementary input tail vanishes by the factorial bound and the uniform conditional second moment of S.

In the physical normalization $C \to J_s^2/\lambda$, define

$$D_N = \sum_a (p - a) h_a, \qquad \mathcal{S}_N = \sum_{a \le b, r} (a + b - r) W_2 Z_{ab;r}^+, \qquad \mathcal{H}_N = \sum_{a \le b, r} g_H W_2 Z_{ab;r}^+. \tag{34}$$

Fixed-window convergence and the preceding tails prove

$$\begin{aligned} \mathcal{H}_N = {} & C[2p(1-q)^2 + K_0] W_2 h^2 \\ & + 8\lambda C(q^2 - q) W_2 \{h, D_N\} + 8\lambda C q^2 \mathcal{S}_N + o_{L^2}(1). \end{aligned} \tag{35}$$

This result is at exact half filling. It does not yet include the weighted A contribution or the full charge-window average.

6 The remaining spectral problem and verification

Equation (35) separates a conserved background, an input-deficit insertion, and an output-deficit insertion. The former input-filter work identifies the centered linear deficit with the existing F direction. The new task is the coherent norm and cross spectrum of $\mathcal{S}_N$ with A and $J_h F$.

At half filling the exact selection rule supplies a useful starting point. For independent normalized pure-rank Hermitian U_a, V_b,

$$P_r(V_b U_a) = (-1)^{a+b-r} P_r(U_a V_b), \tag{36}$$

and therefore

$$\mathbb{E}\|P_r\{U_a, V_b\}\|_m^2 = 4\,\mathbf{1}_{\{a+b-r \ meven\}}\,\pi_{ab}^{(m)}(r). \tag{37}$$

Thus the same-charge symmetric Gram requires parity-weighted deficit moments. The previous first three ordinary moments do not alone determine them. The lower-charge and mixed Gram retain their coherent recoupling data; existing reduced-product formulas [1] explicitly retain multiplicity and adjacent-sector phases.

The A coefficient in Eq. (22) is bounded and has a fixed-deficit limit, but replacing it by that limit under the full operator sum still needs a weighted tail estimate for $A_{ab;r}$. The crude bound on Z^- separately loses the cancellation of its order-m part against Z^+. We have not used that crude bound to claim convergence of the entire memory core. The final raw Z_2 also requires restoring the low-chaos terms of the fourth-chaos descendants before its spectrum can be assembled.

Evidence. Five systems, $(N, p, m) = (4, 1, 2), (5, 2, 3), (6, 2, 3), (6, 2, 4), (6, 3, 4)$, give 522 exact rank-projected checks of the five individual contractions and their core sum on complex, pure-rank inputs. Separately, Eq. (4) is checked as a complete linear map for every external index and output rank: 87 map identities, 5,470,368 complex-entry equalities, maximum cleared integer 60. These exhaustive transfer checks complement the analytic proof; the product checks alone are not a full quadratic-tensor audit.

Symbolic algebra independently merges the five coefficients, derives the finite term and the uniform rational expansions. Twenty-four finite cases use exact binomial fractions and 80/120 digit evaluations, with actual finite-p deviations retained. For $\lambda = 1/2, p = 128, (u, v, S) = (2, 1, 3)$, the g_A/C deviation is about 0.02487; the finite g_H/C deviation is about -0.02144. Precision agreement is not interval certification. All outputs and the six-page compiled note are saved locally, without Git publication.

References

[1] V. K. B. Kota and M. Vyas, *Random Matrix Theory for Transition Strength Densities in Finite Quantum Systems*, Secs. 3, 7, Eqs. (34)–(35), (113), arXiv:1504.00759.